

ORF307

Geometry and Polyhedra

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Midterm 1 Preparations

**Thursday October 9, 10:40 AM -- 12:00 PM,
at Room 120 Lewis Library**

We have uploaded past exams on Canvas/Modules

See announcements on Canvas/EdDiscussion!

At the end of each lecture slide, we have listed relevant chapters from the reference books. *All* exercises within those chapters are excellent practice for Midterm 1. To guide your preparation efforts, we also provide a list of some *representative* exercises from those chapters:

1 Least squares

From the [VMLS] book check the following exercises:

Chapter 12: 12.1, 12.4, 12.10, 12.15

Chapter 13: 13.3, 13.5, 13.7, 13.9

Chapter 15: 15.1, 15.4

Chapter 16: 16.3, 16.4, 16.9, 16.7, 16.10, 16.12

Chapter 17: 17.1, 17.4, 17.6

2 Linear Optimization

From the [LO] book check the following exercises:

Chapter 1: 1.3, 1.4, 1.6, 1.8, 1.10, 1.12, 1.14, 1.15

References

[VMLS]: S. Boyd, L. Vandenberghe: Introduction to Applied Linear Algebra – Vectors, Matrices, and Least Squares;

Available: <https://web.stanford.edu/~boyd/vmls/> .

[LO]: D. Bertsimas & J. Tsitsiklis: Introduction to Linear Optimization.

Available: Princeton Controlled Digital Lending .

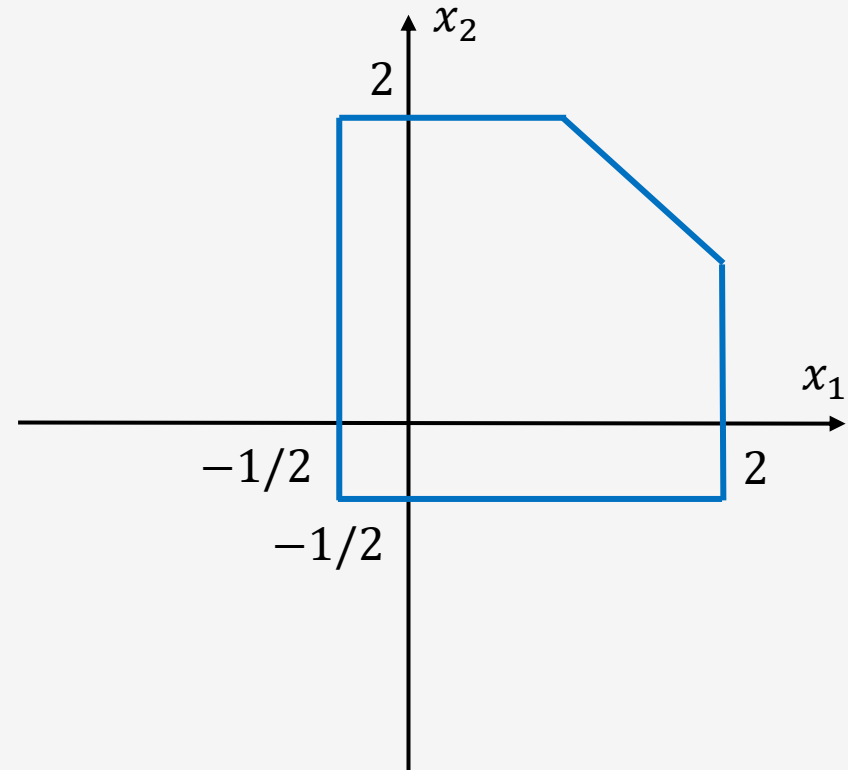
Additional exercises

Today's topics

- Simple example
- Polyhedra
- Corners: extreme points, vertices, basic feasible regions
- Constructing basic solutions
- Existence and optimality of extreme points

A simple example

$$\begin{array}{ll}\min & c^T x \\ \text{s.t.} & -1/2 \leq x_1 \leq 2 \\ & -1/2 \leq x_2 \leq 2 \\ & x_1 + x_2 \leq 3\end{array}$$



**For different objective functions
we get different solutions**

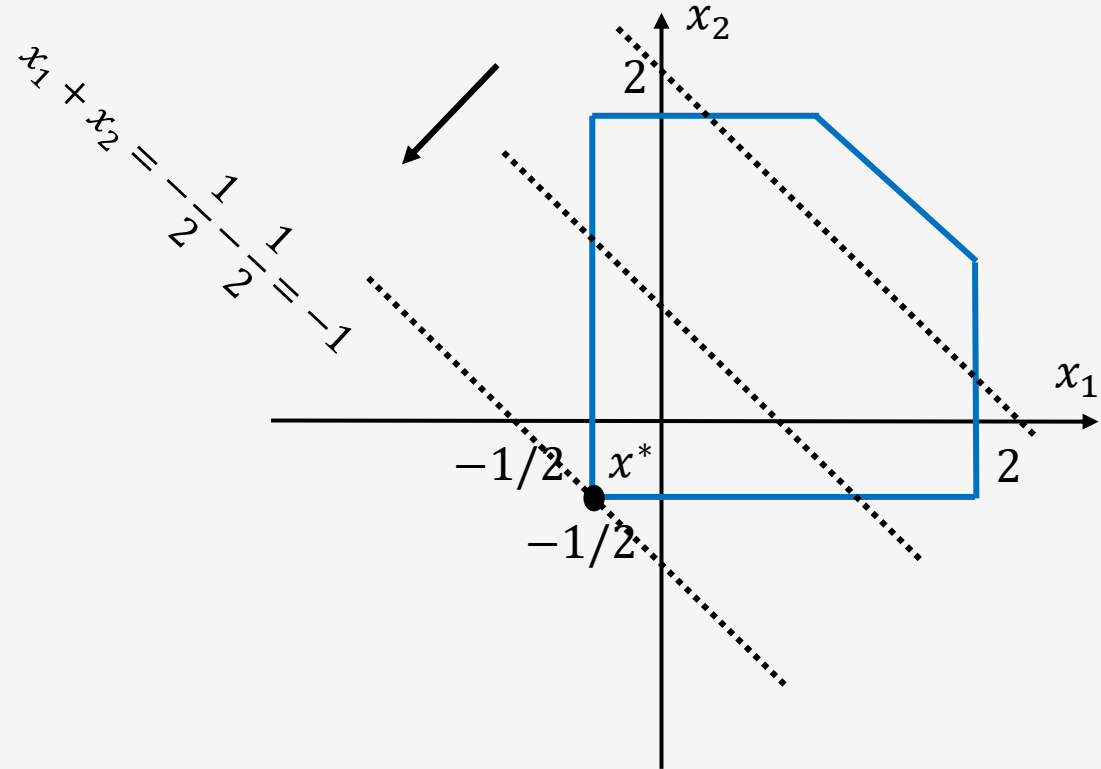
A simple example

Case #1:

$$\begin{array}{ll}\min & c^T x \\ \text{s.t.} & -1/2 \leq x_1 \leq 2 \\ & -1/2 \leq x_2 \leq 2 \\ & x_1 + x_2 \leq 3\end{array}$$

Suppose: $c = (1,1)^T$

$$c^T x = x_1 + x_2$$



Observation: The optimal solution is on a corner or vertex.

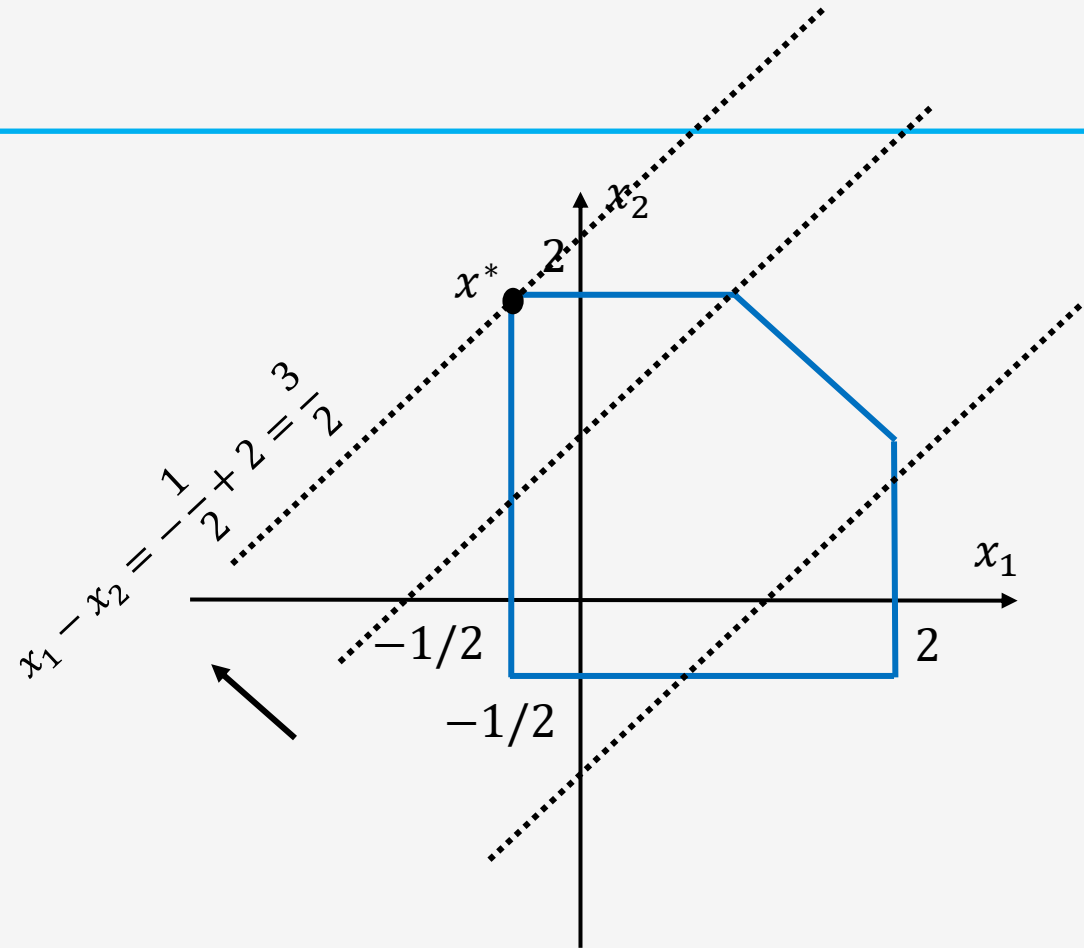
A simple example

Case #2:

$$\begin{array}{ll}\min & c^T x \\ \text{s.t.} & -1/2 \leq x_1 \leq 2 \\ & -1/2 \leq x_2 \leq 2 \\ & x_1 + x_2 \leq 3\end{array}$$

Suppose: $c = (1, -1)$

$$c^T x = x_1 - x_2$$



Observation: The optimal solution is on a corner or vertex.

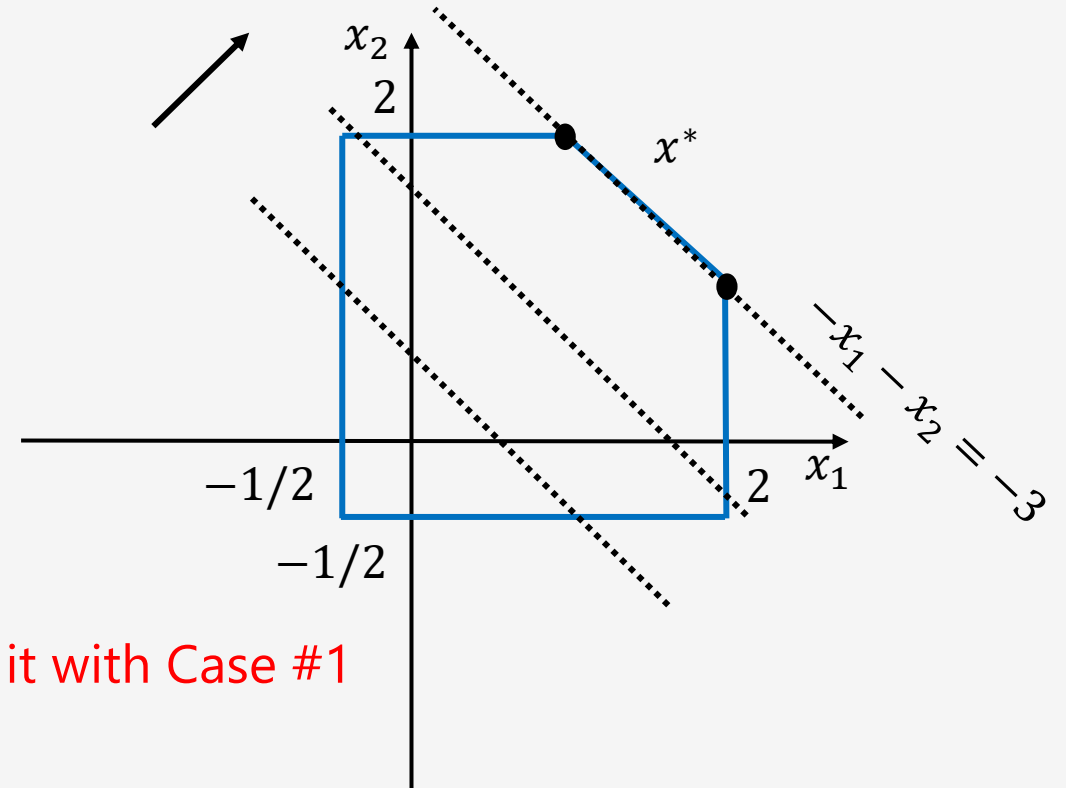
A simple example

Case #3:

$$\begin{aligned} \min \quad & c^T x \\ \text{s.t.} \quad & -1/2 \leq x_1 \leq 1/2 \\ & -1/2 \leq x_2 \leq 1/2 \\ & x_1 + x_2 \leq 3 \end{aligned}$$

Suppose: $c = (-1, -1)$

$$c^T x = -x_1 - x_2$$



Compare it with Case #1

Observation:

- Infinite number of optimal solutions.
- There are still two vertices/corners that are optimal.

-
- **Polyhedra and linear algebra**

Hyperplanes and halfspaces

Hyperplane

$$H = \{x \in \mathbb{R}^n \mid a^T x = b\}$$

$\nabla(a^T x - b) = a$

$-\nabla(a^T x - b) = -a$

$\pi/2$

x_0

x

- x_0 is specific point on the hyperplane H
- For any random point x on the hyperplane H ,
we have $x - x_0 \perp a \Leftrightarrow (x - x_0)^T a = 0$

Halfspace

$$S = \{x \in \mathbb{R}^n \mid a^T x \leq b\}$$

$a^T x = b$

a

x_0

$-a$

S

$a^T x \geq b$

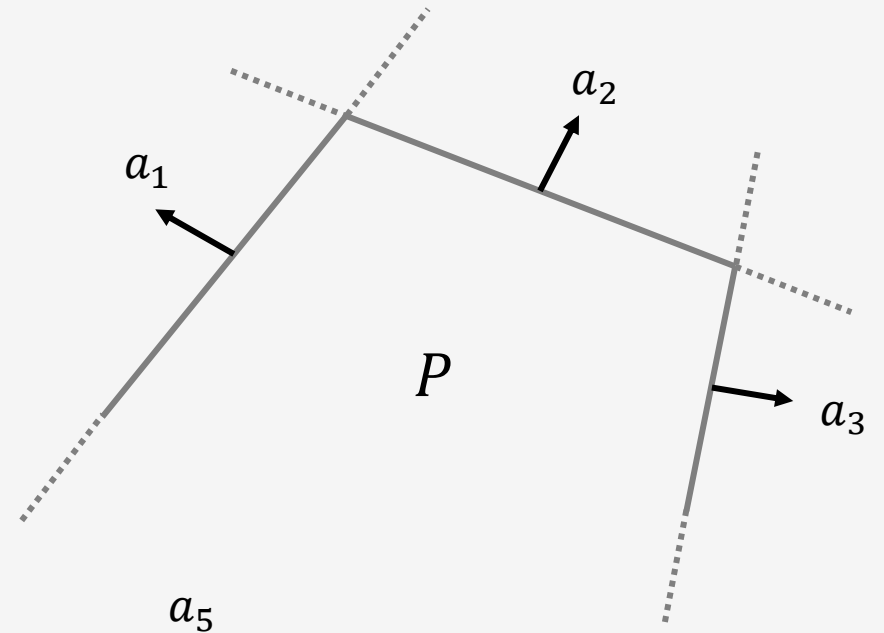
$a^T x \leq b$

- The halfspace S extends in the direction $-a$

Polyhedron

$$P = \{x \in R^n \mid a_i^T x \leq b_i, i = 1, 2, \dots, m\} = \{x \in R^n \mid Ax \leq b\}$$

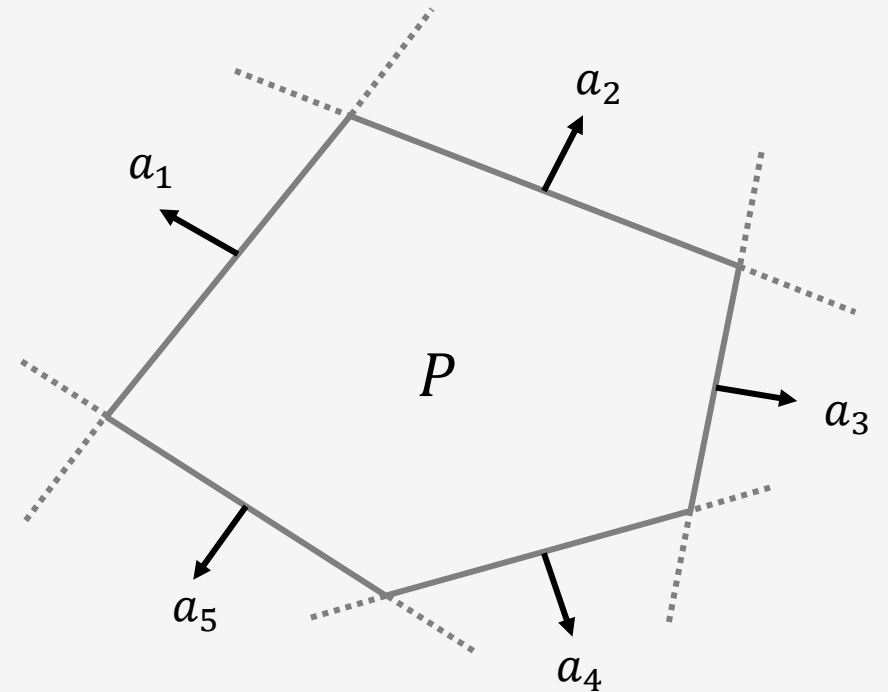
- **Intersection** of a finite number of halfspaces ($\leq$)
- It can also contain hyperplanes ($=$)



Polytope

$$P = \{x \in R^n \mid a_i^T x \leq b_i, i = 1, 2, \dots, m\} = \{x \in R^n \mid Ax \leq b\}$$

- A **polytope** is a **bounded polyhedron**
- **Remark:** A polytope is bounded if it can be enclosed in a ball/sphere.
- **Remark:** many times the terms Polyhedron and polytope are used interchangeably.

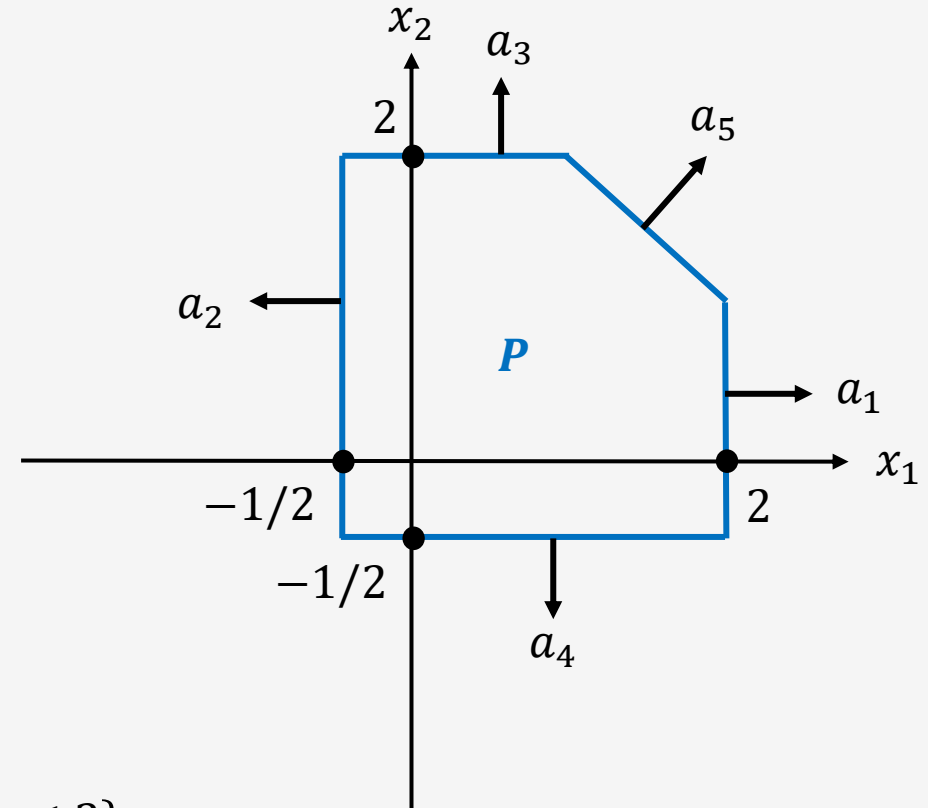


A simple example - *revisited*

$$\begin{array}{ll}\min & c^T x \\ \text{s.t.} & x_1 \leq 2 \\ & -x_1 \leq 1/2 \\ & x_2 \leq 2 \\ & -x_2 \leq 1/2 \\ & x_1 + x_2 \leq 3\end{array}$$

The feasible region is a Polytope (or polyhedron):

$$\mathbf{P} = \{x \in \mathbb{R}^2 \mid x_1 \leq 2, -x_1 \leq 1/2, x_2 \leq 2, -x_2 \leq 1/2, x_1 + x_2 \leq 3\}$$

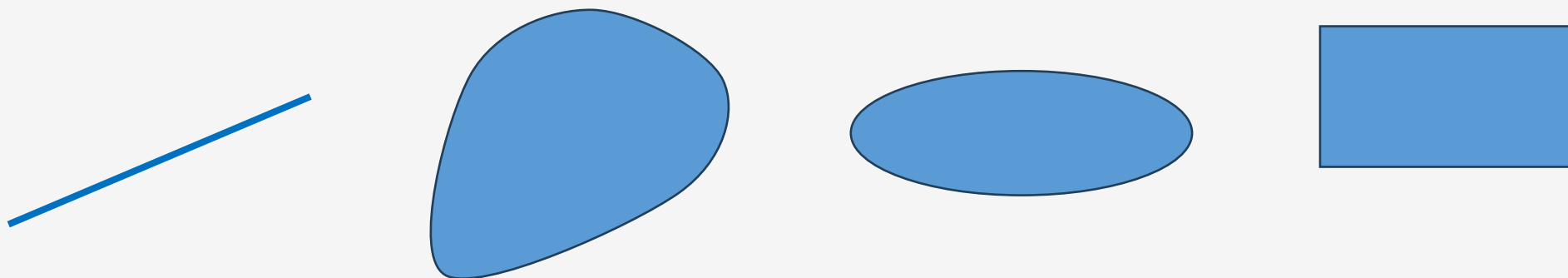


Exercise: determine the vectors a_i for $i = 1, 2, 3, 4, 5$ and justify their directions

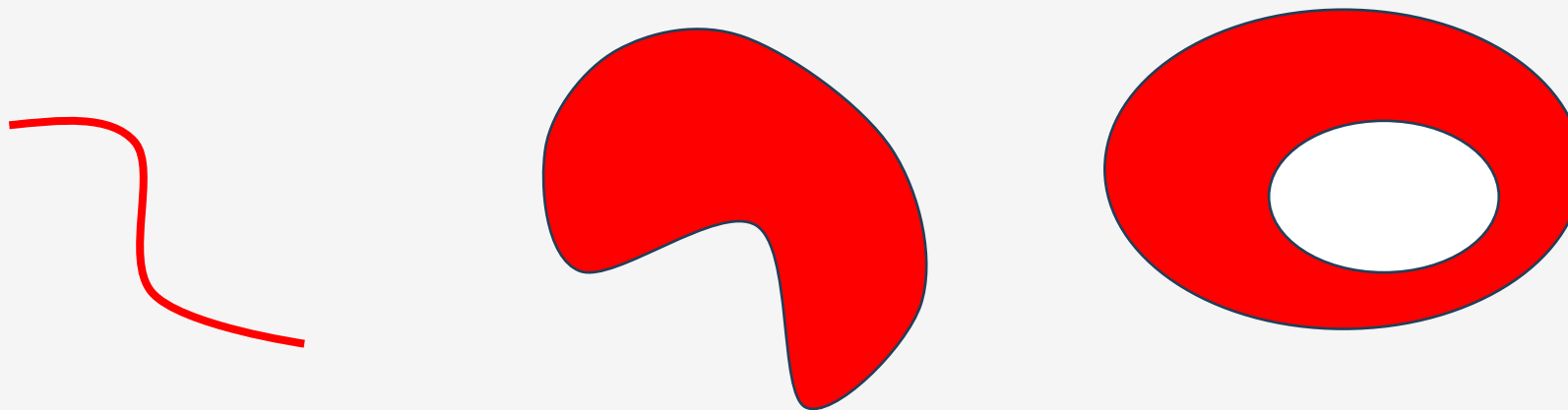
Convex sets

Definition: a set $\mathcal{C}$ is called convex if for any two points $x, y \in \mathcal{C}$ and any $\alpha \in [0, 1]$ the new point $z = \alpha x + (1 - \alpha)y \in \mathcal{C}$

Convex sets



Nonconvex sets



Convex combinations

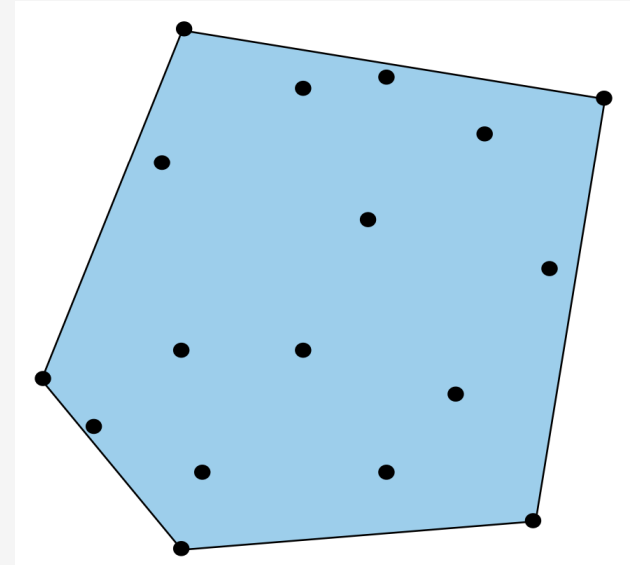
Definition: the point $v \in R^n$ is called the **convex combination** of a finite set of points

$C = \{x_1, x_2, \dots, x_k\}$ there exists $\alpha_i \geq 0, i = 1, \dots, k$ such that

$$v = \alpha_1 x_1 + \dots + \alpha_k x_k \quad \text{and} \quad \alpha_1 + \dots + \alpha_k = 1$$

Definition: the **convex hull** of a set of points $C = \{x_1, x_2, \dots, x_k\}$ is the set of all possible convex combinations of the points in C .

$$\text{conv } C = \left\{ \sum_{i=1}^k \alpha_i x_i \mid \mathbf{1}^T x = 1, \quad \alpha_i \geq 0, i = 1, \dots, k \right\}$$



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- **Corners**

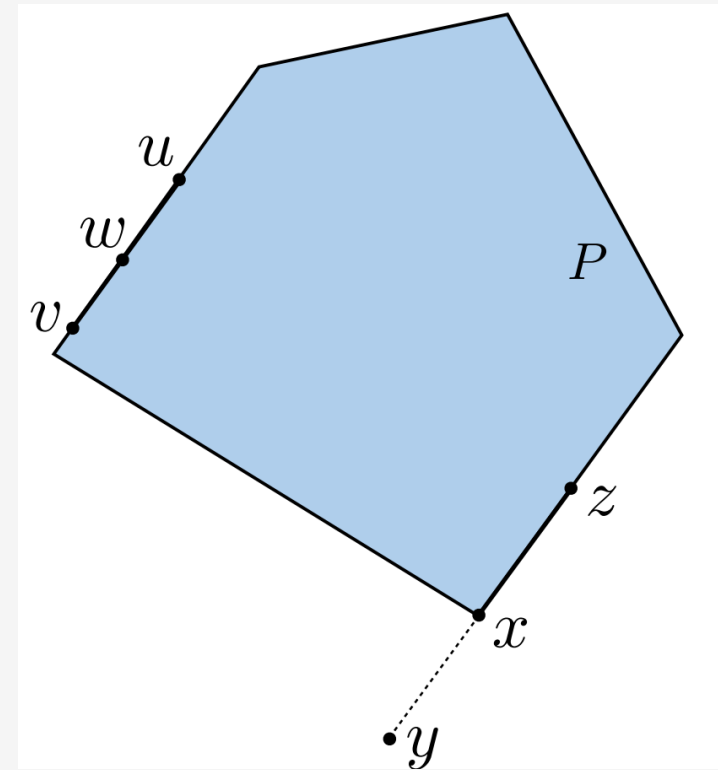
Extreme points

Definition (intuitive): an extreme point of a set is one that is not on a straight line between any two points in the set.

Definition (mathematical): the point $x \in P$ is an extreme point if

$$\nexists y \neq z \in P, \nexists \alpha \in [0,1]: x = \alpha y + (1 - \alpha)z$$

In words: the point $x \in P$ is an extreme point if it cannot be written as convex combination of other different feasible points

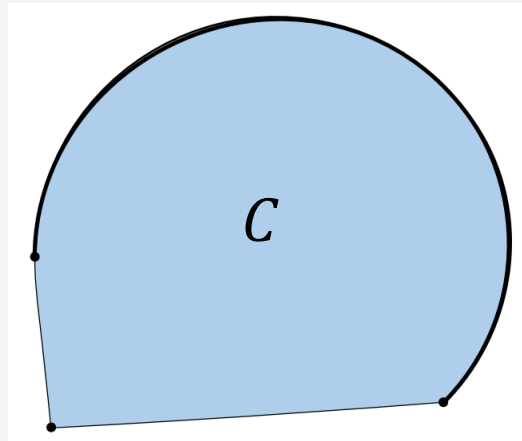


Extreme points

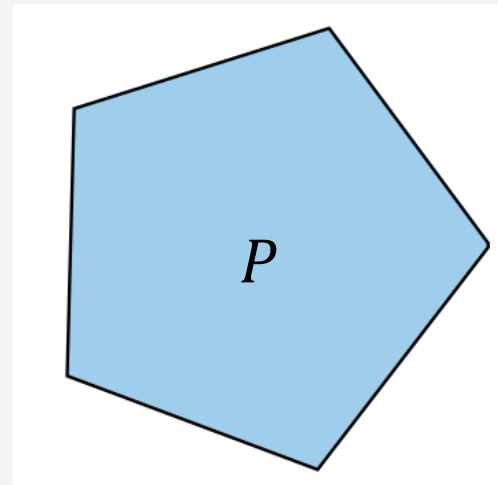
Remark #1: general convex sets can have an infinite number of extreme points

Remark #2: polyhedra are convex sets that have a finite number of extreme points

General convex set



Polyhedron

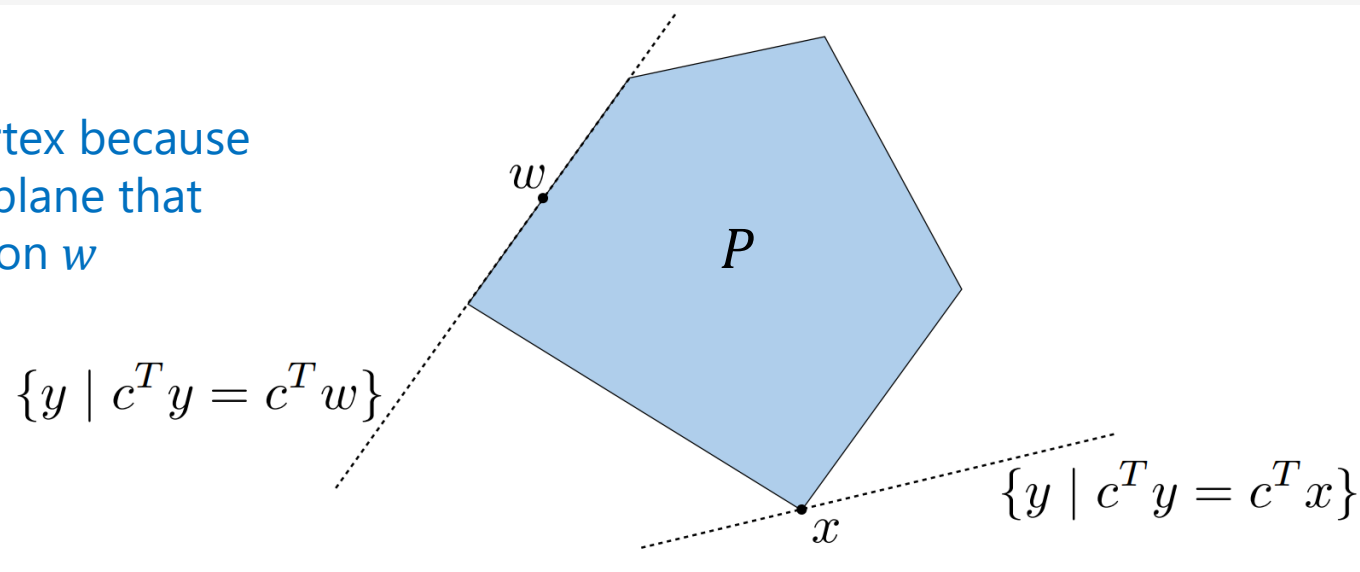


Vertices

Definition: a point $x \in P \subset R^n$ is called a **vertex** if $\exists c \in R^n$ such that x is the **unique** optimal solution of the LP

$$\begin{array}{ll} \min & c^T y \\ \text{s.t.} & y \in P \end{array}$$

$w \in P$ is not a vertex because there is no hyperplane that intersects P only on w



Unique optimal solution:

$$c^T x < c^T y, \forall y \in P,$$

$$x \neq y$$

Basic feasible solutions

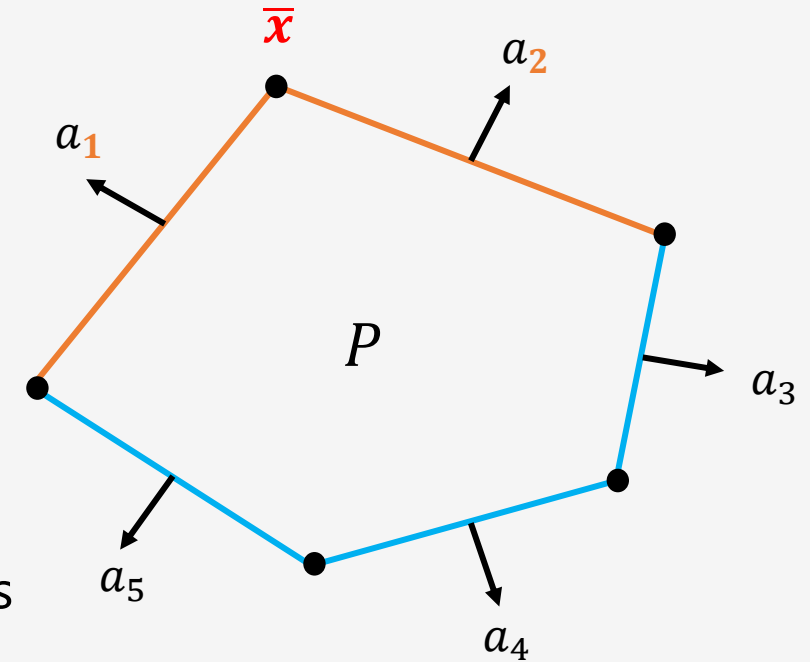
Consider the **polytope**: $P = \{x \in R^n \mid a_i^T x \leq b_i \ i = 1, \dots, m\} \subset R^n$

The set of **active constraints** at $x \in R^n$ is defined by the index set:

$$\mathcal{I}(\bar{x}) = \{i \in \{1, \dots, m\} \mid a_i^T \bar{x} = b_i\}$$

$x \in R^n$ is a **basic solution** if the set $\{a_i \mid i \in \mathcal{I}(\bar{x})\}$ has n linearly independent vectors

$\bar{x} \in P$ is a **basic feasible solution** if the set $\{a_i \mid i \in \mathcal{I}(\bar{x})\}$ has n linearly independent vectors

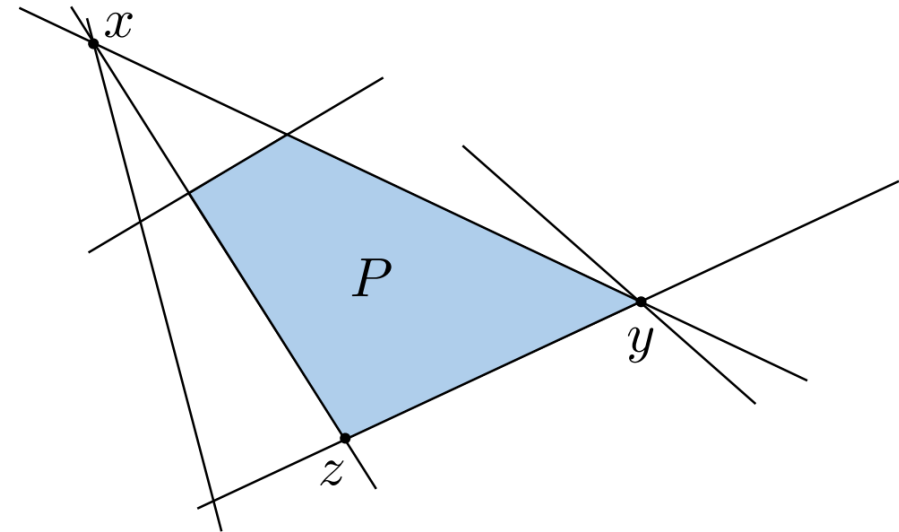


Degenerate basic feasible solutions

Definition: a solution $\hat{x}$ of an LP is called **degenerate** if $|\mathcal{I}(\hat{x})| > n$

True or False?

	Basic	Feasible	Degenerate
x			
y			
z			



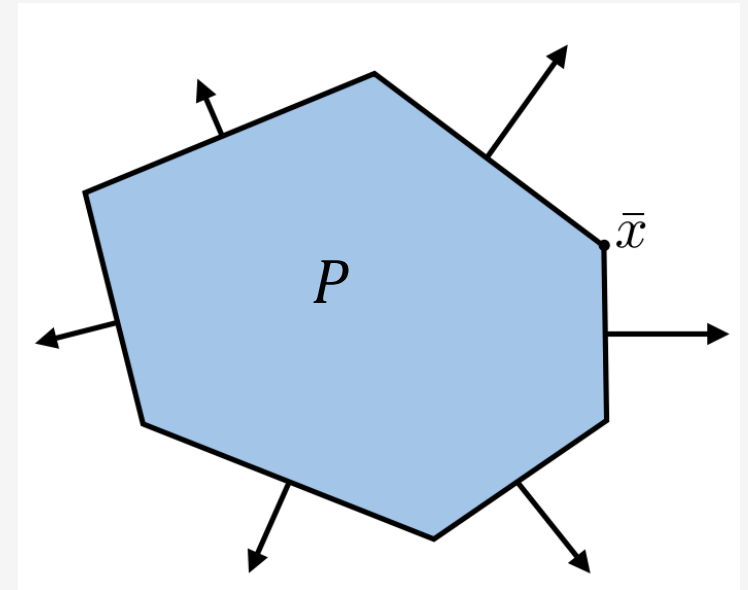
An equivalence theorem

What is the relationship among **vertices**, **extreme points** and **basic solutions** of a **polyhedron**?

$$P = \{x \in R^n \mid a_i^T x \leq b_i \ i = 1, \dots, m\} \subset R^n$$

Theorem: if $\bar{x} \in P$ and P is a polyhedron, then

$\bar{x}$ is a **vertex** $\Leftrightarrow \bar{x}$ is an **extreme point** $\Leftrightarrow \bar{x}$ is a **basic solution**



An equivalence theorem – Proof

Vertex $\Rightarrow$ **Extreme point** (proof by contradiction)

$$\begin{array}{ll} \min & c^T x \\ \text{s.t.} & x \in P \end{array}$$

We know that if x^* is a **vertex** $\Rightarrow \exists c \in R^n$ such that $c^T x^* < c^T y$ for all $y \in P$ (1)

Assume that x^* is **not an extreme point** $\Rightarrow$

$$\exists y, z \in P, \alpha \in [0,1], \text{ with } y, z \neq x^* \text{ such that } x^* = \alpha y + (1 - \alpha)z \quad (2)$$

Since x^* is a **vertex** $\Rightarrow c^T x^* < c^T y$ and $c^T x^* < c^T z$

(2)

(1)

Hence: $c^T x^* = \alpha c^T y + (1 - \alpha)c^T z > \alpha c^T x^* + (1 - \alpha)c^T x^* = c^T x^*$ (**contradiction!**)

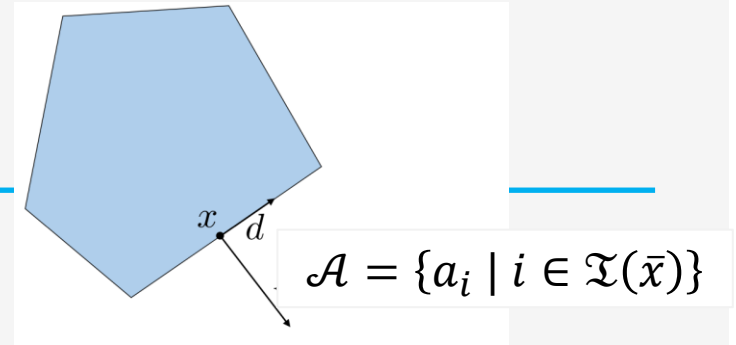
An equivalence theorem – Proof

Extreme point $\Rightarrow$ **Basic feasible solution** (proof by contradiction)

Let $\bar{x} \in P$ be an extreme point.

Assume that $\bar{x}$ is **not a basic feasible solution** $\Rightarrow$ the set $\mathcal{A} = \{a_i \mid i \in \mathcal{I}(\bar{x})\}$ **does not span** R^n .

Which means: $\exists d \in R^n$ that is **perpendicular** to all of them, $a_i^T d = 0, \forall i \in \mathcal{I}(\bar{x})$ (**WHY?**)



(set of active constraints)

(i.e., there are less than n active constraints)

An equivalence theorem – Proof

Extreme point $\Rightarrow$ **Basic feasible solution** (proof by contradiction)

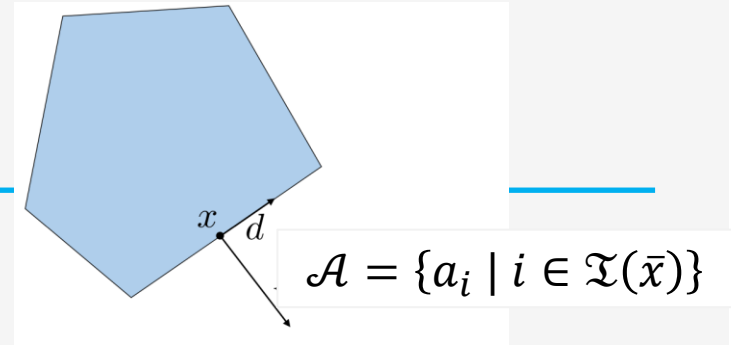
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Consider, $\epsilon > 0$ and define **two new points** $y, z \in R^n$ such that $y = \bar{x} + \epsilon d$ and $z = \bar{x} - \epsilon d$

1. For $i \in \mathcal{I}(\bar{x})$: $a_i^T y = a_i^T \bar{x} + \epsilon a_i^T d = b_i$ and $a_i^T z = a_i^T \bar{x} - \epsilon a_i^T d = b_i$
 2. For $i \notin \mathcal{I}(\bar{x})$: $a_i^T \bar{x} < b_i \Rightarrow a_i^T (\bar{x} + \epsilon d) < b_i$ and $a_i^T (\bar{x} - \epsilon d) < b_i$
- Hence: $y, z \in P$



(set of active constraints)

(i.e., there are less than n active constraints)

An equivalence theorem – Proof

Extreme point $\Rightarrow$ **Basic feasible solution** (proof by contradiction)

Let $\bar{x} \in P$ be an extreme point.

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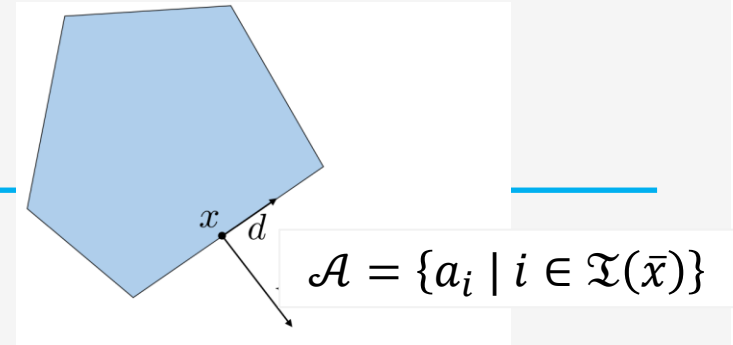
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 2. For $i \notin \mathcal{I}(\bar{x})$: $a_i^T \bar{x} < b_i \Rightarrow a_i^T (\bar{x} + \epsilon d) < b_i$ and $a_i^T (\bar{x} - \epsilon d) < b_i$
- Hence: $y, z \in P$

Take $\lambda = 1/2 \in [0,1]$ and form: $\lambda y + (1 - \lambda)z = \dots = \bar{x}$

This is a **contradiction** because we know that $\bar{x}$ is an extreme point.



An equivalence theorem – Proof

Basic feasible solution $\Rightarrow$ **Vertex** (proof by contradiction)

Exercise for you!

Hint: define $c = \sum_{i \in I(x)} a_i$

See also page 52 in the **[LO]** book

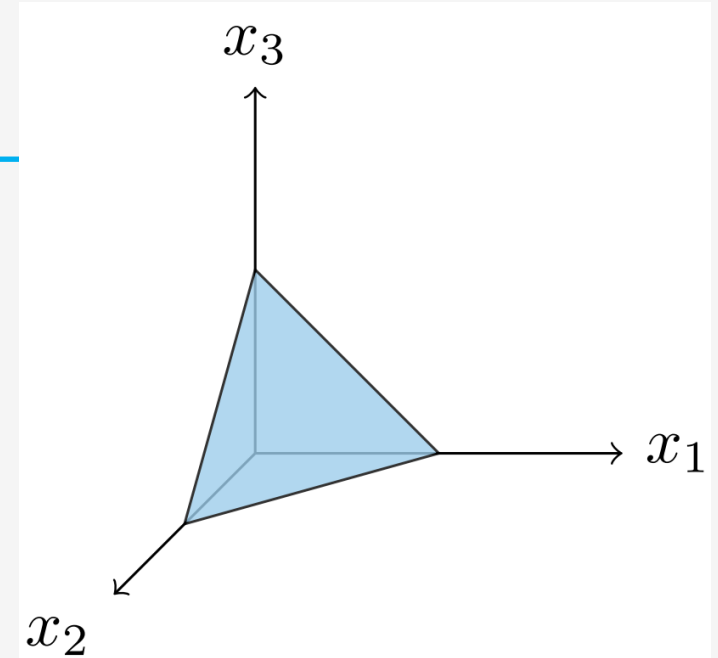
[LO] book: D. Bertsimas & J. Tsitsiklis: *"Introduction to Linear Optimization"*

-
- **Constructing basic feasible solutions**

3D example

Consider the simple LP:

$$\begin{array}{ll}\min & c^T y \\ \text{s.t.} & x_1 + x_2 + x_3 = 1 \\ & x_1, x_2, x_3 \geq 0\end{array}$$



There are $m = 1$ equality constraint and $n = 3$ variables

Basic feasible solution: $\bar{x} \in R^3$ must have $n = 3$ linearly independent active constraints



Only $n - m = 2$ inequalities have to be **active** (i.e., $x_i = 0$, for some $i = 1, 2, 3$)

3D example

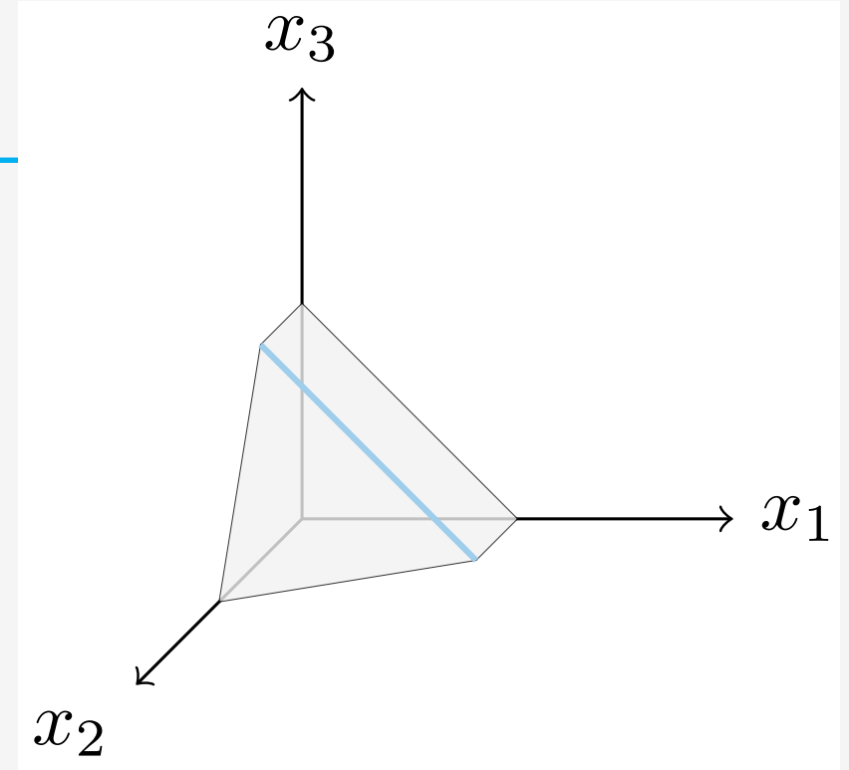
Consider the simple LP:

$$\min \quad c^T y$$

$$\text{s.t.} \quad x_1 + x_3 = 1$$

$$(1/2)x_1 + x_2 + (1/2)x_3 = 1$$

$$x_1, x_2, x_3 \geq 0$$



There are $m = 2$ equality constraints and $n = 3$ variables

Basic feasible solution: $\bar{x} \in R^3$ must have $n = 3$ linearly independent active constraints



Only $n - m = 1$ inequality has to be **active** (i.e., $x_i = 0$, for some $i = 1, 2, 3$)

Constructing a basic feasible solution

$$\min \quad c^T y$$

$$\text{s.t.} \quad x_1 + x_3 = 1$$

$$(1/2)x_1 + x_2 + (1/2)x_3 = 1$$

$$x_1, x_2, x_3 \geq 0$$

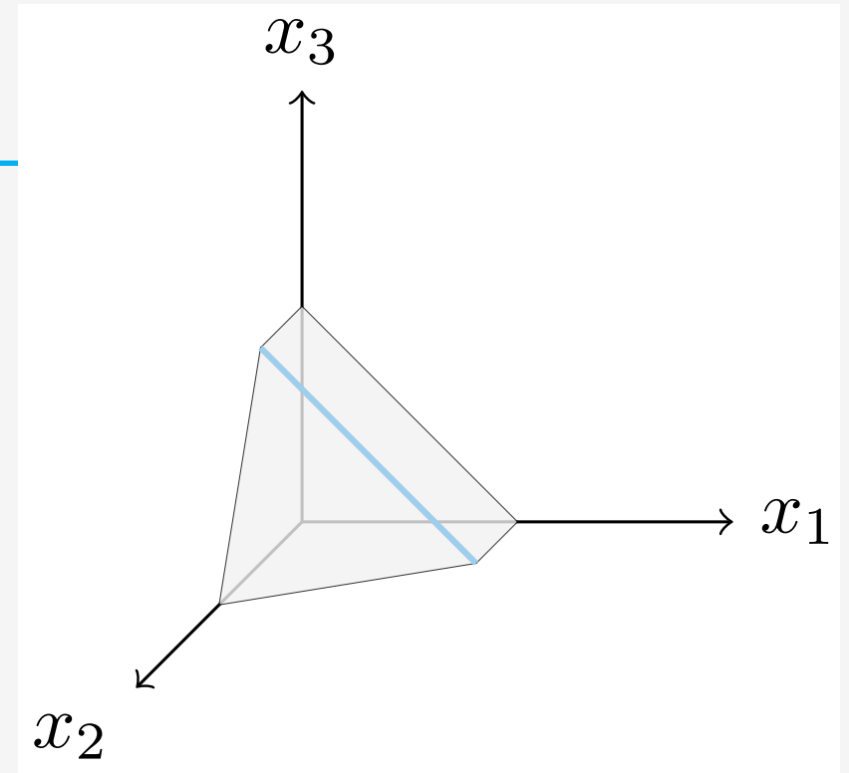
Only $n - m = 1$ inequality has to be **active** (i.e., tight $x_i = 0$, $i = 1, 2, 3$)

You could set $x_1 = 0$ and solve

$$\begin{bmatrix} 1 & 0 & 1 \\ 1/2 & 1 & 1/2 \end{bmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \Rightarrow \begin{bmatrix} 0 & 1 \\ 1 & 1/2 \end{bmatrix} \begin{pmatrix} x_1 \\ x_3 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \Rightarrow \begin{pmatrix} x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 0.5 \\ 1 \end{pmatrix}$$

Since $x_2, x_3 \geq 0$
the point $(0, 0.5, 1)$ is a
basic feasible solution

Remark: Setting $x_2 = 0$ or $x_3 = 0$ you get different basic solutions



3D example

Consider the simple LP:

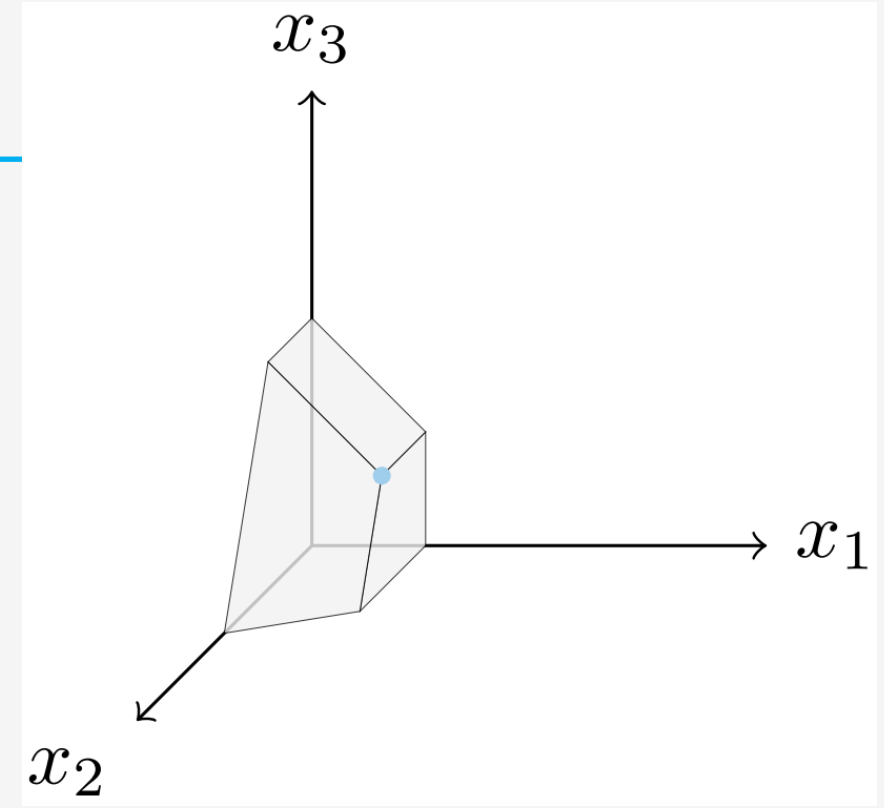
$$\min \quad c^T y$$

$$\text{s.t.} \quad x_1 + x_3 = 1$$

$$(1/2)x_1 + x_2 + (1/2)x_3 = 1$$

$$2x_1 = 1$$

$$x_1, x_2, x_3 \geq 0$$



There are $m = 3$ equality constraints and $n = 3$ variables

Basic feasible solution: $\bar{x}$ must have $n = 3$ linearly independent active constraints



Only $n - m = 0$ inequalities have to be active (i.e., $x_i \neq 0$, $i = 1, 2, 3$)

Standard form polyhedra

Consider the LP in standard form:

$$\begin{array}{ll}\min & c^T y \\ \text{s.t.} & Ax = b \\ & x \geq 0\end{array}$$

Assumptions:

$A \in R^{m \times n}$ has **full row rank** and $m \leq n$
A is a wide matrix

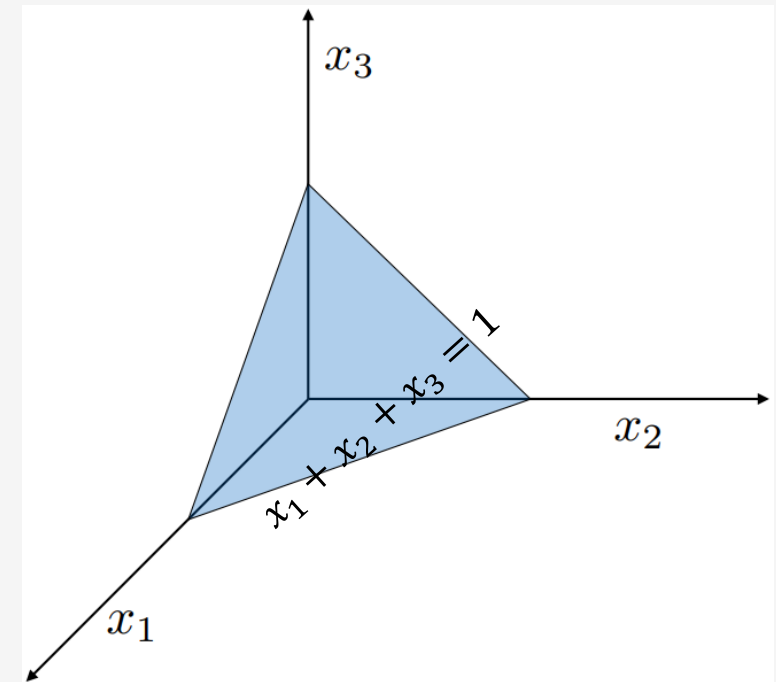
Interpretation:

P is an $(n - m)$ -**dimensional surface**

Or P defines an $(n - m)$ subspace of R^n

The feasible region is a standard polyhedron

$$P = \{x \in R^n \mid Ax = b, x \geq 0\}$$



Example with $m = 1$ and $n = 3$

Basic solutions

Consider the standard form polyhedron

$$P = \{x \in R^n \mid Ax = b, x \geq 0\} \quad \text{with} \quad A \in R^{m \times n} \quad \text{having} \quad \mathbf{full \ row \ rank} \quad (m \leq n)$$

The point $x \in R^n$ is a **basic solution** if and only if

- $Ax = b$
- There exists m indices $B(1), \dots, B(m)$ such that
 - The columns $A_{B(1)}, \dots, A_{B(m)}$ are **linearly independent**
 - And $x_i = 0$ for all $i \neq B(1), \dots, B(m)$

$x \in R^n$ is a **basic feasible solution** if it is **basic solution** and **$x \geq 0$**

Constructing basic solutions

A **basic solution** is computed by following the following steps:

Step #1: Choose any m independent columns of $A \rightarrow A_{B(1)}, \dots, A_{B(m)}$ (basic columns)

Step #2: Set $x_i = 0$ for all $i \neq B(1), \dots, B(m)$

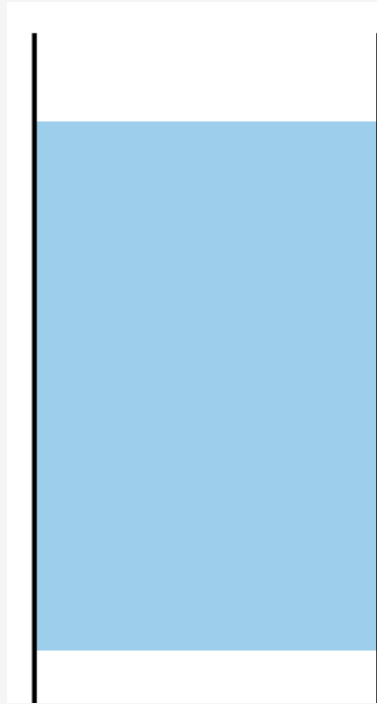
Step #3: solve $Ax = b$ for the remaining variables $\rightarrow x_{B(1)}, \dots, x_{B(m)}$ (basic variables)

$$Ax = b \Leftrightarrow A_B x_B = b \Leftrightarrow \begin{bmatrix} | & & | \\ A_{B(1)} & \dots & A_{B(m)} \\ | & & | \end{bmatrix} \begin{pmatrix} x_{B(1)} \\ \vdots \\ x_{B(m)} \end{pmatrix} = \begin{pmatrix} b_1 \\ \vdots \\ b_n \end{pmatrix}$$

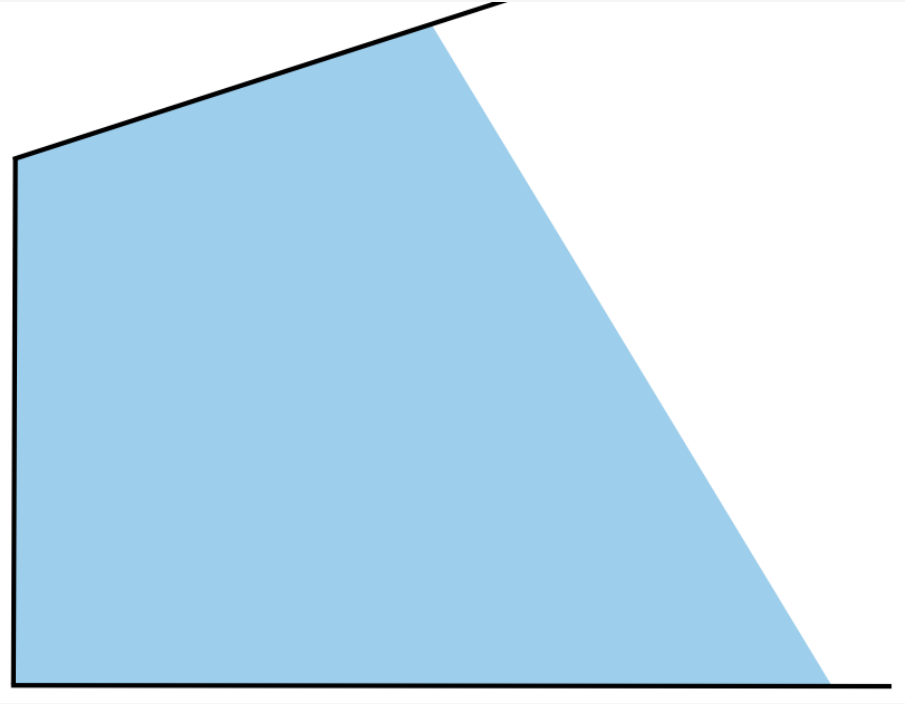
If $x_B \geq 0$ then x is a **basic feasible solution**

Existence and optimality of extreme points

Existence of extreme points



No extreme points
(contains a line)



Extreme points
(does not contain a line)

Existence of extreme points (*characterization*)

A polyhedron P **contains a line** if

$\exists x \in P$ and a nonzero direction $d \in R^n$ such that $x + \lambda d \in P, \forall \lambda \in R$

Given a polyhedron $P = \{x \mid a_i^T x \leq b_i, i = 1 \dots, m\}$, the following are **equivalent**:

P does not contain a line



P has at least one extreme point



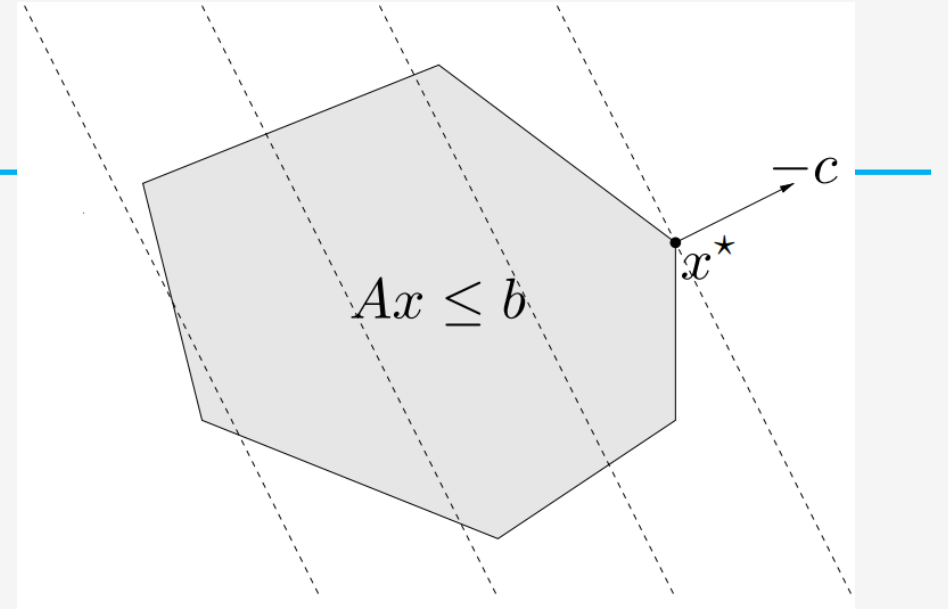
n of the a_i vectors are linearly independent

Corollary: every nonempty bounded polyhedron has at least one basic feasible solution

Optimality of extreme points

Consider the LP: $\min c^T y$
s.t. $Ax \leq b$

and its feasible region: $P = \{x \mid Ax \leq b\}$



Theorem: If P has at least one extreme point and there exists an optimal solution,
then there exists an optimal solution **which** is an extreme point of P .

Solution method: restrict your search for the optimal solution to extreme points only!

How to search among basic feasible solutions

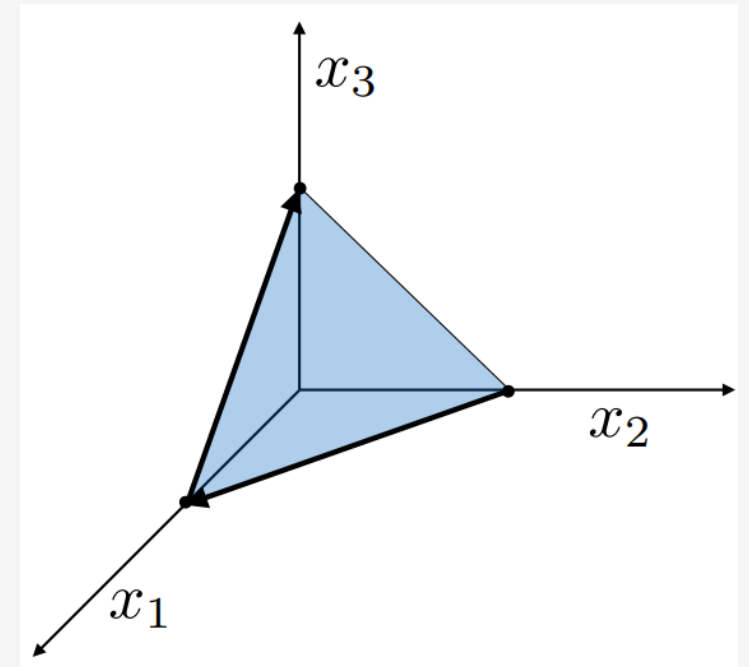
Naïve idea:

Find all basic feasible solutions and select the one that gives the lowest objective value

It is intractable → if $n = 1000$ and $m = 100$ you have to find 10^{143} solutions!

A more efficient idea (~Simplex method)

1. Start at a corner of the feasible region
2. Move to a neighboring corner with lower objective value
3. Go to 2, until there is no further improvement



References

- Bertsimas/Tsitsiklis: Introduction to Linear Optimization
 - Chapter 2.1-2.6: geometry of LP

Next lecture

- More applications
- The Simplex method
